ОПТИМАЛЬНОЕ УПРАВЛЕНИЕ. ДИНАМИЧЕСКОЕ ПРОГРАММИРОВАНИЕ БЕЛЛМАНА ДЛЯ НЕПРЕРЫВНЫХ СИСТЕМ

Постановка задачи

Принцип оптимальности (Р. Беллман, 1957). Каковы бы ни были первоначальные состояния системы и управление в начальный момент, последующее управление должно быть оптимальным относительно состояния, получающегося в результате первого управления

Объект управления: $\dot{x}(t) = f(x(t), u(t))$

Критерий оптимальности: $J = F(x(t_f)) + \int_{t_0}^{t_f} f_0(x(t), u(t))dt$

Функция Беллмана: $S(x(\tau), \tau) = \min_{u(t)} \left[F(x(t_f)) + \int_{\tau}^{t_f} f_0(x(t), u(t))dt \right]$

$S(x(t_f), t_f) = F(x(t_f))$

Решение задачи

Уравнение Беллмана:

$$\min_{u(t)}\left[f_0(x(t),u(t)) + \frac{\partial S}{\partial x}\dot{x} + \frac{\partial S}{\partial t} \right] = 0$$



$$\min_{u(t)}\left[f_0(x(t),u(t)) + \frac{\partial S}{\partial x} f(x(t),u(t)) \right] = -\frac{\partial S}{\partial t}$$

Закон оптимального управления:

$$f_0(x(t),u(t)) + \frac{\partial S}{\partial x} f(x(t),u(t)) = -\frac{\partial S}{\partial t}$$

$$\frac{\partial}{\partial u}\left[f_0(x(t),u(t)) + \frac{\partial S}{\partial x} f(x(t),u(t)) \right] = 0$$

Пример

Объект управления:

$$\dot{x}_1(t) = a_{11}x_1(t) + a_{12}x_2(t) + b_1u(t)$$
$$\dot{x}_2(t) = a_{21}x_1(t) + a_{22}x_2(t) + b_2u(t)$$

Критерий оптимальности:

$$J = \lambda_1 x_1^2(t_f) + \lambda_2 x_2^2(t_f) + \int_0^{t_f} q_1 x_1^2(t) + q_{12}x_1(t)x_2(t) + q_{22}x_2^2(t) + ru^2(t)dt$$

Уравнение Беллмана:

$$\min_{u(t)}\left[q_1x_1^2 + q_{12}x_1x_2 + q_{22}x_2^2 + ru^2 + \frac{\partial S}{\partial x}\dot{x} \right] = -\frac{\partial S}{\partial t}$$



$$q_1x_1^2 + q_{12}x_1x_2 + q_{22}x_2^2 + ru^2 + \frac{\partial S}{\partial x_1}(a_{11}x_1 + a_{12}x_2 + b_1u) + \frac{\partial S}{\partial x_2}(a_{21}x_1 + a_{22}x_2 + b_2u) = -\frac{\partial S}{\partial t}$$

Пример

Закон оптимального управления:

$$q_1 x_1^2 + q_{12} x_1 x_2 + q_{22} x_2^2 + r u^2 + \frac{\partial S}{\partial x_1}(a_{11} x_1 + a_{12} x_2 + b_1 u) + \frac{\partial S}{\partial x_2}(a_{21} x_1 + a_{22} x_2 + b_2 u) = -\frac{\partial S}{\partial t}$$

$$S(x(t_f), t_f) = \lambda_1 x_1^2(t_f) + \lambda_2 x_2^2(t_f)$$

$$\frac{\partial}{\partial u}\left[-\frac{\partial S}{\partial t}\right] = 2ru + \frac{\partial S}{\partial x_1} b_1 + \frac{\partial S}{\partial x_2} b_2 = 0 \Rightarrow u = -\frac{1}{2r}\left(\frac{\partial S}{\partial x_1} b_1 + \frac{\partial S}{\partial x_2} b_2\right)$$

$$\Downarrow$$

$$q_1 x_1^2 + q_{12} x_1 x_2 + q_{22} x_2^2 + r\left(\frac{1}{2r}\left(\frac{\partial S}{\partial x_1} b_1 + \frac{\partial S}{\partial x_2} b_2\right)\right)^2 + \frac{\partial S}{\partial x_1}\left(a_{11} x_1 + a_{12} x_2 - \frac{b_1}{2r}\left(\frac{\partial S}{\partial x_1} b_1 + \frac{\partial S}{\partial x_2} b_2\right)\right)$$

$$+ \frac{\partial S}{\partial x_2}\left(a_{21} x_1 + a_{22} x_2 - \frac{b_2}{2r}\left(\frac{\partial S}{\partial x_1} b_1 + \frac{\partial S}{\partial x_2} b_2\right)\right) = -\frac{\partial S}{\partial t}$$

Пример

Закон оптимального управления:

$$q_1 x_1^2 + q_{12} x_1 x_2 + q_{22} x_2^2 + r\left(\frac{1}{2r}\left(\frac{\partial S}{\partial x_1} b_1 + \frac{\partial S}{\partial x_2} b_2\right)\right)^2 + \frac{\partial S}{\partial x_1}\left(a_{11} x_1 + a_{12} x_2 - \frac{b_1}{2r}\left(\frac{\partial S}{\partial x_1} b_1 + \frac{\partial S}{\partial x_2} b_2\right)\right)$$

$$+ \frac{\partial S}{\partial x_2}\left(a_{21} x_1 + a_{22} x_2 - \frac{b_2}{2r}\left(\frac{\partial S}{\partial x_1} b_1 + \frac{\partial S}{\partial x_2} b_2\right)\right) = -\frac{\partial S}{\partial t}$$



$$q_1 x_1^2 + q_{12} x_1 x_2 + q_{22} x_2^2 + \frac{1}{4r}\left(\frac{\partial S}{\partial x_1} b_1\right)^2 + \frac{1}{2r}\frac{\partial S}{\partial x_1}\frac{\partial S}{\partial x_2} b_1 b_2 + \frac{1}{4r}\left(\frac{\partial S}{\partial x_2} b_2\right)^2 + \frac{\partial S}{\partial x_1} a_{11} x_1 + \frac{\partial S}{\partial x_1} a_{12} x_2$$

$$- \frac{1}{2r}\left(\frac{\partial S}{\partial x_1} b_1\right)^2 - \frac{b_1}{2r}\frac{\partial S}{\partial x_1}\frac{\partial S}{\partial x_2} b_2 + \frac{\partial S}{\partial x_2} a_{21} x_1 + \frac{\partial S}{\partial x_2} a_{22} x_2 - \frac{b_2}{2r}\frac{\partial S}{\partial x_2}\frac{\partial S}{\partial x_1} b_1 - \frac{1}{2r}\left(\frac{\partial S}{\partial x_2} b_2\right)^2 = -\frac{\partial S}{\partial t}$$

Пример

Закон оптимального управления:

$$q_1 x_1^2 + q_{12} x_1 x_2 + q_{22} x_2^2 + \frac{1}{4r}\left(\frac{\partial S}{\partial x_1} b_1\right)^2 + \frac{1}{2r}\frac{\partial S}{\partial x_1}\frac{\partial S}{\partial x_2} b_1 b_2 + \frac{1}{4r}\left(\frac{\partial S}{\partial x_2} b_2\right)^2 + \frac{\partial S}{\partial x_1} a_{11} x_1 + \frac{\partial S}{\partial x_1} a_{12} x_2$$

$$-\frac{1}{2r}\left(\frac{\partial S}{\partial x_1} b_1\right)^2 - \frac{b_1}{2r}\frac{\partial S}{\partial x_1}\frac{\partial S}{\partial x_2} b_2 + \frac{\partial S}{\partial x_2} a_{21} x_1 + \frac{\partial S}{\partial x_2} a_{22} x_2 - \frac{b_2}{2r}\frac{\partial S}{\partial x_2}\frac{\partial S}{\partial x_1} b_1 - \frac{1}{2r}\left(\frac{\partial S}{\partial x_2} b_2\right)^2 = -\frac{\partial S}{\partial t}$$



$$q_1 x_1^2 + q_{12} x_1 x_2 + q_{22} x_2^2 - \frac{1}{4r}\left(\frac{\partial S}{\partial x_1} b_1\right)^2 - \frac{b_2}{2r}\frac{\partial S}{\partial x_2}\frac{\partial S}{\partial x_1} b_1 - \frac{1}{4r}\left(\frac{\partial S}{\partial x_2} b_2\right)^2 +$$

$$+\frac{\partial S}{\partial x_1} a_{11} x_1 + \frac{\partial S}{\partial x_1} a_{12} x_2 + \frac{\partial S}{\partial x_2} a_{21} x_1 + \frac{\partial S}{\partial x_2} a_{22} x_2 = -\frac{\partial S}{\partial t}$$

Пример

Закон оптимального управления:

$$q_1 x_1^2 + q_{12} x_1 x_2 + q_{22} x_2^2 - \frac{1}{4r}\left(\frac{\partial S}{\partial x_1} b_1\right)^2 - \frac{b_2}{2r}\frac{\partial S}{\partial x_2}\frac{\partial S}{\partial x_1} b_1 - \frac{1}{4r}\left(\frac{\partial S}{\partial x_2} b_2\right)^2 +$$

$$+ \frac{\partial S}{\partial x_1} a_{11} x_1 + \frac{\partial S}{\partial x_1} a_{12} x_2 + \frac{\partial S}{\partial x_2} a_{21} x_1 + \frac{\partial S}{\partial x_2} a_{22} x_2 = -\frac{\partial S}{\partial t}$$

$$q_1 x_1^2 + q_{12} x_1 x_2 + q_{22} x_2^2 - \left(\frac{1}{2\sqrt{r}}\left(\frac{\partial S}{\partial x_1} b_1\right) + \frac{1}{2\sqrt{r}}\left(\frac{\partial S}{\partial x_2} b_2\right)\right)^2 + \frac{\partial S}{\partial x_1}(a_{11} x_1 + a_{12} x_2) + \frac{\partial S}{\partial x_2}(a_{21} x_1 + a_{22} x_2) = -\frac{\partial S}{\partial t}$$

Выбор функции Беллмана:

$$S = \psi_1 x_1^2 + \psi_{12} x_1 x_2 + \psi_2 x_2^2$$

Пример

$$q_1 x_1^2 + q_{12} x_1 x_2 + q_{22} x_2^2 - \frac{1}{4r}\left(b_1\left(2\psi_1 x_1 + \psi_{12} x_2\right) + b_2\left(\psi_{12} x_1 + 2\psi_2 x_2\right)\right)^2 +$$

$$+(2\psi_1 x_1 + \psi_{12} x_2)(a_{11} x_1 + a_{12} x_2) + (\psi_{12} x_1 + 2\psi_2 x_2)(a_{21} x_1 + a_{22} x_2) = -(\dot{\psi}_1 x_1^2 + \dot{\psi}_{12} x_1 x_2 + \dot{\psi}_2 x_2^2)$$

$$\left(q_1 + 2a_{11}\psi_1 + a_{21}\psi_{12} - \left(\frac{b_1\psi_1}{\sqrt{r}}\right)^2 - \left(\frac{b_2\psi_{12}}{2\sqrt{r}}\right)^2 - \frac{b_1 b_2 \psi_1 \psi_{12}}{r}\right) x_1^2 +$$

$$+\left(q_{12} + 2a_{12}\psi_1 + 2a_{21}\psi_2 + a_{11}\psi_{12} + a_{22}\psi_{12} - \frac{b_1 b_2 \psi_{12}^2}{2r} - \frac{b_1^2 \psi_1 \psi_{12}}{r} - \frac{b_2^2 \psi_2 \psi_{12}}{r} - \frac{2b_1 b_2 \psi_1 \psi_2}{r}\right) x_1 x_2$$

$$+\left(q_{22} + 2a_{22}\psi_2 + a_{12}\psi_{12} - \left(\frac{b_2\psi_2}{\sqrt{r}}\right)^2 - \left(\frac{b_1\psi_{12}}{2\sqrt{r}}\right)^2 - \frac{b_1 b_2 \psi_2 \psi_{12}}{r}\right) x_2^2$$

$$= -(\dot{\psi}_1 x_1^2 + \dot{\psi}_{12} x_1 x_2 + \dot{\psi}_2 x_2^2)$$

$$S(x(t_f), t_f) = \lambda_1 x_1^2(t_f) + \lambda_2 x_2^2(t_f) \quad \Rightarrow \quad \psi_1(t_f) = \lambda_1,\ \psi_{12}(t_f) = 0,\ \psi_2(t_f) = \lambda_2$$

Пример

$$\begin{cases} -\dot{\psi}_1 = q_1 + 2a_{11}\psi_1 + a_{21}\psi_{12} - \left(\dfrac{b_1\psi_1}{\sqrt{r}}\right)^2 - \left(\dfrac{b_2\psi_{12}}{2\sqrt{r}}\right)^2 - \dfrac{b_1b_2\psi_1\psi_{12}}{r} \\ -\dot{\psi}_{12} = q_{12} + 2a_{12}\psi_1 + 2a_{21}\psi_2 + a_{11}\psi_{12} + a_{22}\psi_{12} - \dfrac{b_1b_2\psi_{12}^2}{2r} - \dfrac{b_1^2\psi_1\psi_{12}}{r} - \dfrac{b_2^2\psi_2\psi_{12}}{r} - \dfrac{2b_1b_2\psi_1\psi_2}{r} \\ -\dot{\psi}_2 = q_{22} + 2a_{22}\psi_2 + a_{12}\psi_{12} - \left(\dfrac{b_2\psi_2}{\sqrt{r}}\right)^2 - \left(\dfrac{b_1\psi_{12}}{2\sqrt{r}}\right)^2 - \dfrac{b_1b_2\psi_2\psi_{12}}{r} \end{cases}$$

$$\begin{cases} \psi_1(t_f) = \lambda_1 \\ \psi_{12}(t_f) = 0 \\ \psi_2(t_f) = \lambda_2 \end{cases}$$

$$u = -\frac{1}{2r}\left((2\psi_1 x_1 + \psi_{12}x_2)b_1 + (\psi_{12}x_1 + 2\psi_2 x_2)b_2\right)$$

Постановка задачи

Объект управления: $\dot{x}(t) = f(x(t), u(t))$

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Уравнение Беллмана: $\min_{u(t)} \left[f_0(x(t), u(t)) + \frac{\partial S}{\partial x} f(x(t), u(t)) \right] = 0$

Закон оптимального управления: $f_0(x(t), u(t)) + \frac{\partial S}{\partial x} f(x(t), u(t)) = 0$

$$\frac{\partial}{\partial u} \left[f_0(x(t), u(t)) + \frac{\partial S}{\partial x} f(x(t), u(t)) \right] = 0$$

Пример

$$0 = q_1 + 2a_{11}\psi_1 + a_{21}\psi_{12} - \left(\frac{b_1\psi_1}{\sqrt{r}}\right)^2 - \left(\frac{b_2\psi_{12}}{2\sqrt{r}}\right)^2 - \frac{b_1 b_2 \psi_1 \psi_{12}}{r}$$

$$0 = q_{12} + 2a_{12}\psi_1 + 2a_{21}\psi_2 + a_{11}\psi_{12} + a_{22}\psi_{12} - \frac{b_1 b_2 \psi_{12}^2}{2r} - \frac{b_1^2 \psi_1 \psi_{12}}{r} - \frac{b_2^2 \psi_2 \psi_{12}}{r} - \frac{2b_1 b_2 \psi_1 \psi_2}{r}$$

$$0 = q_{22} + 2a_{22}\psi_2 + a_{12}\psi_{12} - \left(\frac{b_2\psi_2}{\sqrt{r}}\right)^2 - \left(\frac{b_1\psi_{12}}{2\sqrt{r}}\right)^2 - \frac{b_1 b_2 \psi_2 \psi_{12}}{r}$$

$$u = -\frac{1}{2r}\left((2\psi_1 x_1 + \psi_{12} x_2)b_1 + (\psi_{12} x_1 + 2\psi_2 x_2)b_2\right)$$